Project Initialization and Planning Phase

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	3 Marks

Project Title

Personalized Travel Itinerary Planner

Problem Statement

Planning a personalized travel itinerary is a time-consuming and overwhelming process. Travelers face significant challenges when trying to organize multi-day trips that align with their budget, interests, group size, and preferred travel style. Existing solutions either offer generic, one-size-fits-all itineraries or require extensive manual research across multiple platforms.

Key Pain Points

1. Information Overload

- Travelers must visit dozens of websites and apps to research destinations, attractions, restaurants, and accommodations
- No single platform consolidates all planning aspects into one coherent day-by-day itinerary
- Difficulty in discovering hidden gems and lesser-known local experiences

2. Budget Management Challenges

- Estimating total trip costs across activities, meals, stays, and transport is tedious
- Hard to keep track of per-day spending limits relative to overall trip budget
- Currency conversion and regional pricing differences add complexity

3. Personalization Gap

- Most travel planning tools provide generic recommendations that don't account for individual preferences
- Group travel planning is especially difficult — balancing interests of multiple travelers
- Travel style (relaxed vs. packed) is rarely factored into itinerary suggestions

4. Itinerary Rigidity

- Traditional itineraries are static and cannot be easily modified once created
- No ability to regenerate or swap individual days while preserving the rest of the plan
- Drag-and-drop reordering of activities is not available in most planners

5. Technical Barriers

- Non-technical users struggle with complex travel planning tools
- Lack of AI-powered assistance for intelligent itinerary generation
- No real-time collaboration features for group trip planning

Target Users

User Type	Description	Pain Point
Solo Travelers	Individual travelers planning personal trips	Information overload, budget tracking
Group Travelers	Friends/families planning trips together	Coordinating preferences, shared budget management
Budget Travelers	Cost-conscious travelers	Need strict budget adherence in planning
Adventure Seekers	Travelers seeking packed, activity-rich itineraries	Finding enough activities, optimal scheduling
Relaxation Seekers	Travelers preferring leisurely, relaxed trips	Avoiding over-scheduling, quality over quantity

Problem Scope

The project aims to solve these problems by building an AI-powered travel itinerary planner that:

- Generates personalized, day-by-day itineraries using Generative AI (Gemini / Groq LLMs)
- Manages budgets with per-day and per-activity cost estimation
- Supports multiple travel styles (relaxed, balanced, packed)
- Allows real-time editing, reordering, and regeneration of individual days
- Provides a modern, responsive web interface with secure user authentication